

Module 10: Relational Data Model & Integrity Constraints

1. Relation Anatomy & Constraints Architecture

RELATIONAL DATA MODEL & INTEGRITY CONSTRAINTS

Relation Anatomy: Tuples, Attributes, Domains, Degree, Cardinality & The 4 Constraints

RELATION ANATOMY: STUDENT(RollNo, Name, Marks)

RollNo (PK)	Name	Marks
101	Akhil Sharma	92
102	Rahul Gupta	85
103	Pooja Verma	95

- Degree (Arity): Number of attributes / columns = 3
- Cardinality: Number of tuples / rows = 3
- Domain: Permitted set of atomic values (e.g. Marks $\in [0, 100]$)

Update Operations & Constraint Checks:

- INSERT: Can violate Domain, Key, Entity, or Referential integrity.
- DELETE: Can only violate Referential Integrity (if child references it).
- UPDATE: Combination of Delete old + Insert new.

DBMS automatically rejects transactions violating constraints!

THE 4 RELATIONAL INTEGRITY CONSTRAINTS

1. Domain Constraint

Every attribute value must be atomic and within defined data type.
Example: Age cannot be "ABC" or a set of values.

2. Key Constraint

No two distinct tuples can have identical candidate key values.
 $t1[Candidate_Key] \neq t2[Candidate_Key]$

3. Entity Integrity Constraint

Primary Key attributes can NEVER be NULL!
If PK is NULL, individual tuples cannot be identified or indexed.
 $t[Primary_Key] \neq NULL$

4. Referential Integrity Constraint

Foreign key value in referencing relation must either:
(a) Match an existing PK in referenced relation, OR (b) Be NULL.
Violation Triggers: ON DELETE CASCADE / SET NULL / RESTRICT

2. The 4 Fundamental Integrity Constraints

Constraint	Formal Rule	Critical Rule
Domain Constraint	Attribute value $\in \text{dom}(A)$	Must be atomic (indivisible)
Key Constraint	$t1[K] \neq t2[K]$ for distinct tuples	Uniqueness across all tuples
Entity Integrity	$t[\text{Primary_Key}] \neq \text{NULL}$	Primary key is NEVER NULL
Referential Integrity	Foreign Key $\rightarrow$ Parent Primary Key	Must match valid parent PK OR be NULL

3. Referential Action Triggers

- CASCADE: Auto-deletes child rows upon parent deletion.
- SET NULL: Sets child FK column to NULL.
- RESTRICT: Rejects parent modification if referenced by active child tuples.